BENNETT UNIVERSITY (THE TIMES GROUP), GREATER NOIDA

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND TECHNOLOGY (SCSET)

Project Report on

"Customer Complaint Sentiment Analysis and Category Analysis"

Submitted in AI ML Project Course, CSET 312

Bachelor of Technology

MILESTONE 2 REPORT

Model Implementation, System Architecture, and Preliminary Results

Team Member	Enrollment No.
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1. Executive Summary & Project Overview

Modern financial services, banking institutions, and fintech enterprises handle high daily volumes of customer complaints. Manual inspection and routing of grievances lead to operational bottlenecks, delayed responses to urgent fraud incidents, and reduced customer satisfaction. The primary objective of this project is to build an end-to-end **AI-based Customer Complaint Intelligence System** that automates text preprocessing, multi-class issue categorization, urgency priority estimation, and targeted operational department routing.

During **Milestone 2**, the team successfully transitioned from initial problem formulation to a fully functional, integrated application. The project artifacts include a validated NLP text preprocessing pipeline, a TF-IDF feature extraction engine, a trained Logistic Regression baseline model, an automated rule-assisted priority escalation module, a persistent SQLite database schema, a FastAPI RESTful backend API, and an interactive Next.js 16 web interface supporting role-based portal access and dark/light theme switching.

2. System Architecture & Conceptual Workflow

The end-to-end data and operational flow of the system follows a structured 7-stage architectural pipeline:

- **1. Intake Portal:** Customers submit unstructured natural language complaint narratives via the web client.
- **2. NLP Preprocessing:** Text cleaning, lowercasing, punctuation stripping, and token normalization.
- **3. Vectorization:** Sublinear term frequency TF-IDF encoding converting text into numerical feature vectors.
- **4. AI Categorization:** Trained Logistic Regression classifier predicts the specific intent/category.
- **5. Priority & Department Routing:** Urgency engine calculates priority (High/Medium/Low) based on intent and emergency flags, then assigns the ticket to specialized departments.
- **6. Database Persistence:** Records are stored in a SQLite relational database with real-time status tracking.
- **7. Admin Intelligence Dashboard:** Support teams filter, monitor, update ticket statuses, and analyze priority distributions.

Pipeline Stage	Component	Technical Execution / Output
Intake & API	FastAPI / Next.js	REST endpoint POST /api/complaints accepting JSON payload
NLP Feature Pipeline	Scikit-Learn TF-IDF	Serialized tfidf_vectorizer.pkl (5,000 max features)
Machine Learning	Logistic Regression	Serialized logistic_regression_model.pkl model
Rule Engine	Priority Evaluator	Flags unauthorized charges & stolen cards as HIGH priority
Data Storage	SQLite DB	complaints.db schema storing tickets & audit timestamps

3. Dataset Description & Preprocessing Methodology

Dataset Sources: The project utilizes real-world consumer complaints sourced from the *Consumer Financial Protection Bureau (CFPB)* dataset and fine-grained intent classes from the *BANKING77* benchmark dataset. The combined dataset spans 77 detailed intent classes covering major financial product domains:

- **Card & Payment Services:** Card loading, unauthorized fees, PIN change, stolen cards, expired cards.
- **Account Management:** Account closure, pending top-up, beneficiary additions, statement errors.
- **Transfers & Exchange:** Delayed international transfers, exchange rate queries, failed SWIFT/UPI debits.
- **Loans & Mortgages:** Interest rate disputes, mortgage payments, credit report discrepancies.

Text Preprocessing Pipeline: Raw complaint narratives undergo rigorous text cleaning:

1. *Case Normalization:* Conversion of all text tokens to lowercase.
2. *Noise & Special Character Removal:* Regex stripping of HTML tags, punctuation, special symbols, and extra whitespace.
3. *Stopword Filtering:* Removal of non-informative English stop words to emphasize discriminative terms.
4. *TF-IDF Feature Extraction:* Term Frequency-Inverse Document Frequency scaling with sublinear term frequency enabled, bi-gram range (1,2), and feature dimension capped at 5,000 top terms.

4. Model Implementation & Preliminary Results

For Milestone 2, a multi-class **Logistic Regression model** with L2 regularization (`solver='lbfgs'`, `C=1.0`) was implemented as the core baseline classifier. The model was evaluated using standard multi-class classification metrics on an 80/20 train-test split:

Evaluation Metric	Score / Value	Description & Analytical Notes
Classification Accuracy	84.2%	Overall multi-class accuracy across 77 banking categories
Macro Precision	0.851	High precision reducing false positive department dispatches
Macro Recall	0.838	Strong recall ensuring critical complaints are captured
Macro F1-Score	0.844	Balanced performance score across balanced and imbalanced intents
Inference Latency	< 45 ms	Ultra-low latency inference ideal for real-time web intake

Case Study Validation: Sample complaints evaluated by the pipeline produced high-confidence dispatches:

- *Sample 1:* 'Someone used my credit card without permission at a merchant and $450 was deducted.' → Predicted Category: **card_payment_not_recognized** | Priority: **HIGH** | Department: **Fraud & Security**.
- *Sample 2:* 'I tried to transfer money yesterday but transaction is pending and funds were debited.' → Predicted Category: **transfer_timing** | Priority: **MEDIUM** | Department: **Payments & Transfers**.

5. Milestone 2 Completion Summary & Team Contributions

All planned deliverables for **Milestone 2** have been completed successfully. The summary of completed milestones and specific team contributions is detailed below:

Team Member	Enrollment No.	Milestone 2 Core Responsibilities & Deliverables
Pranjal Chaturvedi	S24CSEU2082	Data preprocessing pipeline implementation, noise stripping regex, TF-IDF vectorizer configuration, multi-class Logistic Regression model training, and priority rule logic integration.
Tanmay Awal	S24CSEU2098	System architecture design, FastAPI REST API development (`backend/main.py`), SQLite DB schema & seeding, Next.js 16 UI frontend implementation, and project report compilation.

6. Future Scope & Milestone 3 Roadmap

Building upon the successful Milestone 2 deployment, the planned roadmap for Milestone 3 includes:

- **Deep Learning Models:** Fine-tuning DistilBERT and BERT-base transformer models to compare semantic representation against the TF-IDF baseline.
- **Enhanced Analytics:** Integrating Recharts visual charts in the Admin Dashboard for category breakdown and real-time SLA metrics.
- **Automated Escalations:** Email notification webhooks for High-Priority fraud complaints and automated resolution feedback loops.